

Decoupled Data Synthesis Pipeline

The pipeline consists of the following steps:

- Input:** Vidprom (video prompts) and Style Ref (style reference images).
- Filtering:** Both inputs are processed by a **Filter** step.
- Prompt Rewriting:** The filtered Vidprom is processed by a **Rewrite** step, resulting in **Static/Dynamic Prompts**.
- Image Generation:** The filtered Style Ref and the rewritten prompts are fed into **Qwen-Image**.
- Stylized Images:** Qwen-Image generates **stylized images**.
- Stylized Videos:** The stylized images are used to create **stylized videos**.
- Multi-Criteria Scoring:** The stylized videos are evaluated using a **Multi-Criteria Scoring** module.
- Output Datasets:**
 - SFT Dataset (Selected Filtered Text):** Contains the **Best Score** video and its associated **"text"**.
 - Preference Dataset:** Contains **chosen** and **rejected** video-text pairs for preference learning.

Hybrid Image-Video Training

The diagram illustrates the architecture for training a model on both images and videos. It starts with two input types: **Video** and **Image**. Both inputs are processed through a **random-crop** operation to extract patches, which are then arranged in a 4x4 grid. The **Video** path also includes a **Style Image latent** input. These patches are fed into a **Patch Embedding** block. The **Image** path also includes a **"text"** input, which is processed by a **umT5** block. The outputs of the Patch Embedding and the umT5 block are combined via **Cross Attention** and fed into **Nx DiT Blocks** (with a flame icon). A callout provides a detailed view of the **Gated Attention** mechanism, showing multiple heads ($H_1, H_2, \dots, H_n$) and gates ($G_1, G_2, \dots, G_n$) that are concatenated and processed. The final output is **Predicted noise**.

DPO for Post-Training

The diagram illustrates the DPO (Direct Preference Optimization) process for post-training. It shows two input sequences: a "chosen" sequence (top) and a "rejected" sequence (bottom), each represented by a film strip icon and a corresponding image. These sequences are fed into two models: a "Reference model" (top) and a "Policy model" (bottom). The "Reference model" is associated with a snowflake icon, and the "Policy model" is associated with a flame icon. Both models output to a central "DPO" block, which is represented by a large orange rounded rectangle containing a cube icon surrounded by three circles. Arrows indicate the flow of data from the input sequences to the models and then to the DPO block.